

Problem_4: The following Table gives the family budget data for a few samples of **four** low-income classes of families of country for a year.

	Yearly income per consumer unit in taka			
	< 60000	60000 & < 75000	75000 & < 105000	>105000
Number of households in the sample	136	179	111	22
Average number of consumer units per household	2.60	2.57	2.50	2.48
Average income per consumer unit	54300.1	68100.3	86100.9	123200.0
Average expenditure on food per consumer unit	29100.8	33100.6	37400.4	40700.1

Calculate the income-elasticity of demand for food assuming that the demand function has constant elasticity. Also, test the significance of this estimated income-elasticity at 5% level.

Solution: The demand function with constant income-elasticity is given by

$$\log E_f = \alpha + \beta \log E_0$$

where E_f is the expenditure per consumer unit on food and E_0 is the income per consumer unit. The parameters α and β are estimated by the method of least squares, viz, minimizing.

$$\sum_{i=1}^4 n_i (\log E_{f_i} - \alpha - \beta \log E_{0_i})^2$$

where n_i is the number of households at the i^{th} income level.

Let $Y_i = \log E_{f_i}$ and $X_i = \log E_{0_i}$

The weighted least square estimates of α and β are given by

$$\hat{\beta} = \frac{(\sum n_i)(\sum n_i X_i Y_i) - (\sum n_i X_i)(\sum n_i Y_i)}{(\sum n_i)(\sum n_i X_i^2) - (\sum n_i X_i)^2}$$

$$\hat{\alpha} = \frac{(\sum n_i Y_i) - \hat{\beta}(\sum n_i X_i)}{\sum n_i}$$

The income-elasticity of demand is $\eta = \frac{d \log E_f}{d \log E_0} = \hat{\beta} = ???$ which is **less than / greater than** unity. Thus, the demand for food is **inelastic/ elastic** with respect to income.

$$Var(\hat{\beta}) = \frac{\sum n_i}{(\sum n_i)(\sum n_i X_i^2) - (\sum n_i X_i)^2}$$

$$SE(\hat{\beta}) = \sqrt{Var(\hat{\beta})}$$

$Z = \frac{\hat{\beta}}{SE(\hat{\beta})}$ is approximately distributed as N(0,1)

n_i	E_{0i}	E_{fi}	$X_i = \ln(E_{0i})$	$Y_i = \ln(E_{fi})$
136	54300.1	29100.8		
179	68100.3	33100.6		
111	86100.9	37400.4		
22	123200.0	40700.1		

$$\hat{\beta} = \frac{\sum n_i (\sum n_i X_i Y_i) - (\sum n_i X_i)(\sum n_i Y_i)}{(\sum n_i)(\sum n_i X_i^2) - (\sum n_i X_i)^2} = ???$$

$$\hat{\alpha} = \frac{(\sum n_i Y_i) - \hat{\beta}(\sum n_i X_i)}{\sum n_i} = ???$$

$$Var(\hat{\beta}) = \frac{\sum n_i}{(\sum n_i)(\sum n_i X_i^2) - (\sum n_i X_i)^2} = ???$$

$$SE(\hat{\beta}) = \sqrt{Var(\hat{\beta})} = ???$$

Now we want to test

$$H_0: \beta = 0 \quad \text{VS} \quad H_1: \beta \neq 0$$

The test statistic is $Z = \frac{\hat{\beta} - \beta}{SE(\hat{\beta})} \sim N(0,1)$

Under H_0 : $Z = \frac{\hat{\beta}}{SE(\hat{\beta})} = ???$

Comment:

The tabulated value of |z| at 5% level is 1.96